

Literature Status for Question Q:UC of arXiv:1510.00276

FA/Banach run packet

June 28, 2026

Status

Result type: literature already answered.

Source question: Hytönen–Li–Naor, *Quantitative affine approximation for UMD targets*, arXiv:1510.00276, Question Q:UC in Section 1, “Open questions.”

Supporting answer: Hytönen–Naor, *Heat flow and quantitative differentiation*, arXiv:1608.01915, Theorem 1.

Source Question

The source paper asks whether the improved lower bound for the modulus of uniform affine approximability, proved there for UMD targets, remains valid when the target is merely uniformly convex. In its notation, Question Q:UC asks for an improved bound as in Theorem 1 under the weaker assumption that Y admits an equivalent uniformly convex norm, and notes that this is the maximal qualitative generality by the theorem of Bates–Johnson–Lindenstrauss–Preiss–Schechtman.

Supporting Literature Answer

The later paper arXiv:1608.01915 states the following theorem in its introduction.

Theorem. *Suppose that Y admits an equivalent uniformly convex norm. Then there is $c = c(Y) \in (0, \infty)$ such that for every $n \in \mathbb{N}$, every n -dimensional normed space X , and every $\varepsilon \in (0, 1/2]$,*

$$r^{X \rightarrow Y}(\varepsilon) \geq \exp\left(-\frac{1}{\varepsilon^{cn}}\right).$$

Immediately after the theorem, the later paper states that this result answers Question 8 of arXiv:1510.00276 positively. This is a direct supporting-paper identification rather than an agent-inferred implication.

Scope and Limitations

This packet records the positive literature answer to the uniformly convex-target part of Question Q:UC. It does not settle the broader problem of the sharp asymptotic rate of $r_n(\varepsilon)$ in the Hilbert-to-Hilbert case, nor the separate infinite-dimensional-domain question from the same source section.

The supporting paper also improves the Hilbertian L_2 Dorronsoro estimate, but it still records the quantitative differentiation asymptotic problem as open in the sense that the best known Hilbert-to-Hilbert bound for $r^{\ell_2^n \rightarrow \ell_2}(1/4)$ has the same asymptotic form as the general bound.

Search Bounds

Before packaging, the run’s registry, solutions index, attempts index, and proof-gaps index were searched for arXiv:1510.00276, the paper title, “affine approximation,” “quantitative differentiation,” and “UMD targets.” No prior packet for arXiv:1510.00276 was found. The nearby arXiv:1608.01915 registry hit concerns Question 19 / Q:proj lip of the later paper, not this source question.

References

References

- [1] T. Hytönen, S. Li, and A. Naor, *Quantitative affine approximation for UMD targets*, Discrete Analysis 2016:37, 1–71. arXiv:1510.00276. <https://arxiv.org/abs/1510.00276>
- [2] T. Hytönen and A. Naor, *Heat flow and quantitative differentiation*, J. Eur. Math. Soc. 20 (2018), no. 9, 2183–2247. arXiv:1608.01915. <https://arxiv.org/abs/1608.01915>